

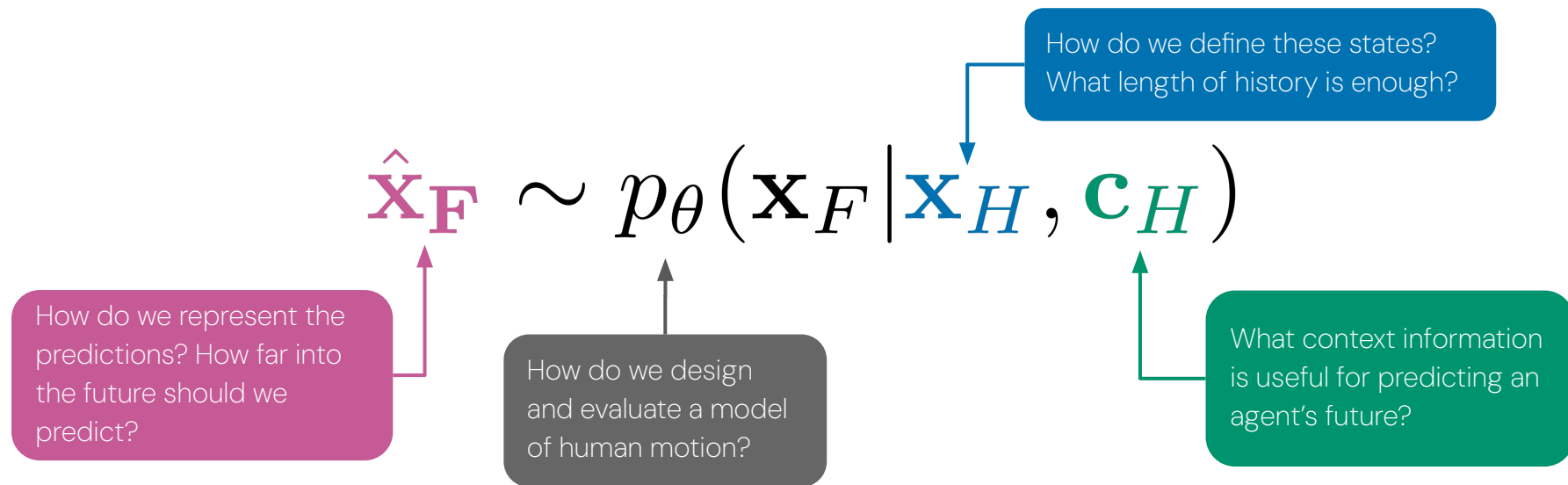
Trajectory Prediction: Taxonomy and Open Challenges

Ingrid Navarro

What is Motion Prediction?

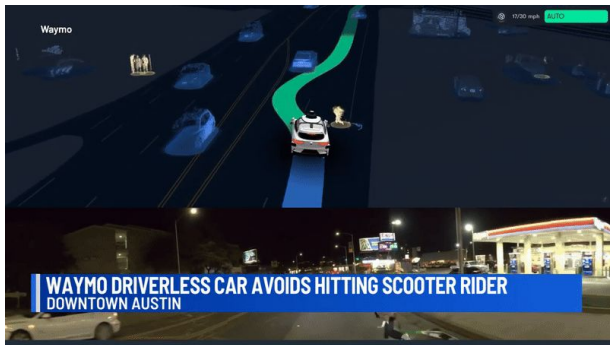
Motion Prediction

The task of forecasting the **future** states of an agent or set of agents given a **history** of states and the **context** surrounding an agent,



Why do we care about motion prediction in robotics?

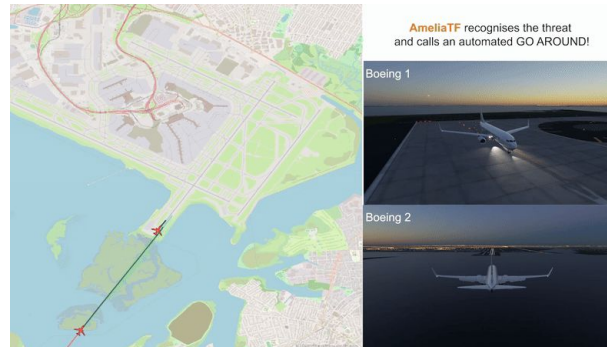
Motion prediction is important because it helps us understand how humans interact in shared spaces, anticipate their future actions, and use that knowledge to guide safe and effective decision-making and decision-support systems.



Source: KXAN



Source: giphy.com



Source: Amelia



Perception



Prediction



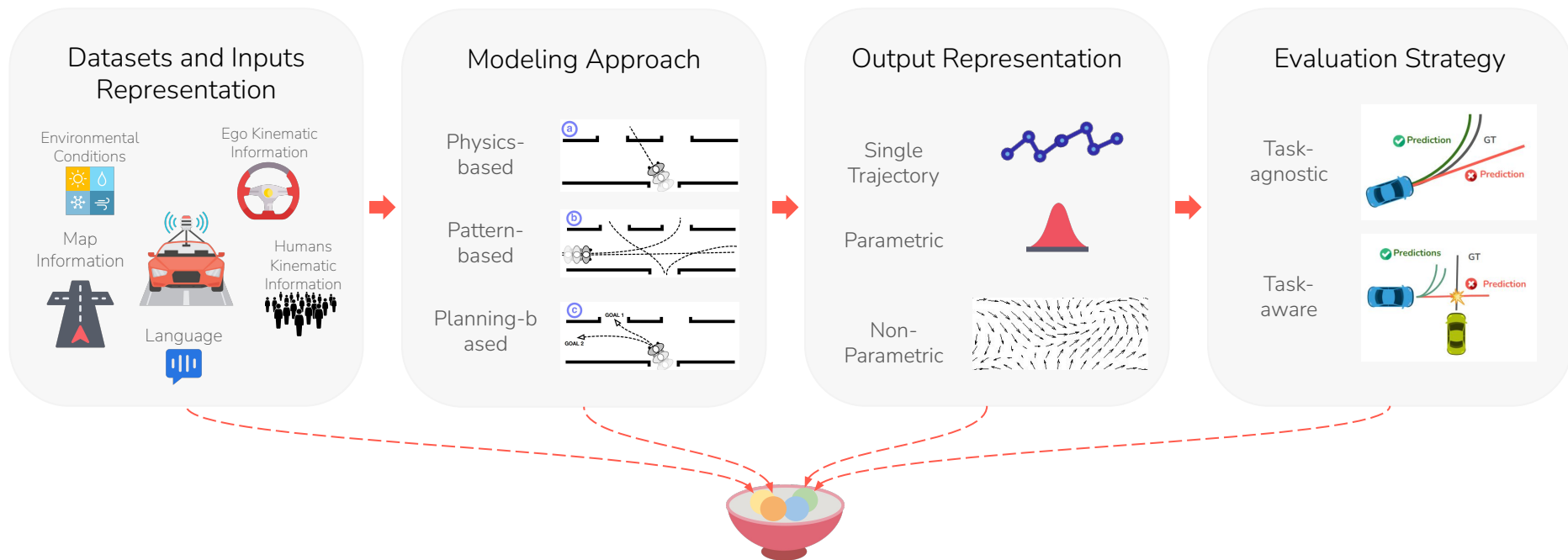
Planning



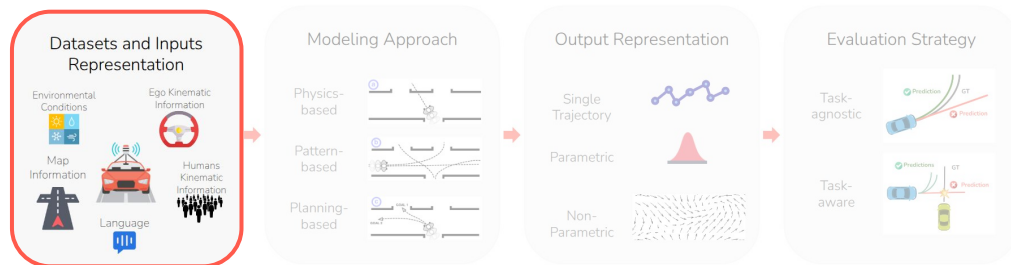
Control



Key Ingredients



Motion Domains and Datasets



Common domains and use cases

Human Motion

Use cases: service robots, sport analysis.



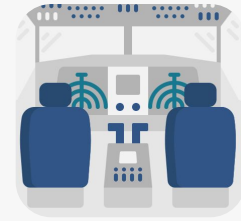
Driving

Use cases: personal self-driving, freight, traffic management.



Aviation

Use cases: traffic management, decision-support systems for task load reduction.



Human Motion



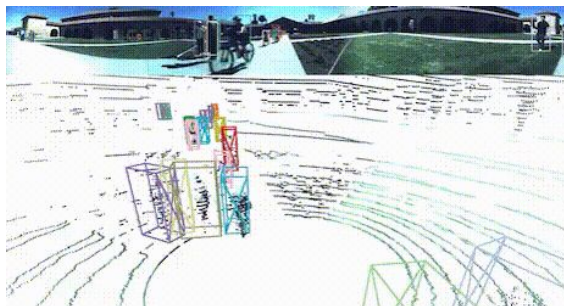
ETH[1] / UCY[2]



Subdomain: Surveillance

Captures human behavior in public spaces, like hotels, universities and stores.

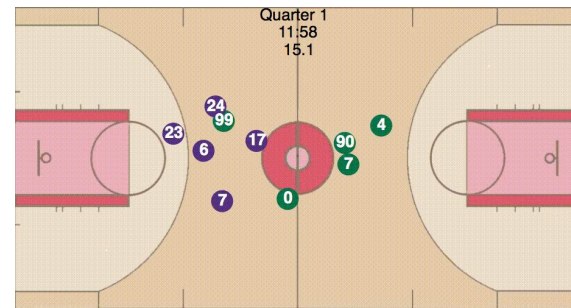
JRDB [3]



Subdomain: Service Robots (Social Navigation)

Captures human behavior in public spaces from a first-person-view.

SportVU Data [4]



Subdomain: Sports

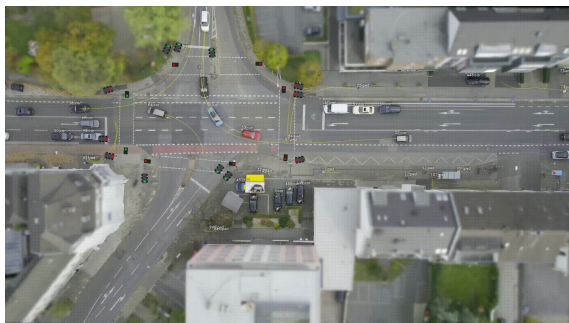
Utilizes a motion capture system that tracks players, balls, referees.

- [1] Lerner, A., Chrysanthou, Y., & Lischinski, D. (2007, September). *Crowds by example*. In *Computer graphics forum* (Vol. 26, No. 3, pp. 655-664). Oxford, UK: Blackwell Publishing Ltd.
- [2] Pellegrini, S., Ess, A., Schindler, K., & Van Gool, L. (2009, September). *You'll never walk alone: Modeling social behavior for multi-target tracking*. In 2009 IEEE 12th international conference on computer vision (pp. 261-268). IEEE.
- [3] Martin-Martin, R., Patel, M., Rezaatoughi, H., Shenoi, A., Gwak, J., Frankel, E., ... & Savarese, S. (2021). *Jrdb: A dataset and benchmark of egocentric robot visual perception of humans in built environments*. *IEEE transactions on pattern analysis and machine intelligence*, 45(6), 6748-6765.
- [4] *SportsVU Data*: <https://github.com/linouk23/NBA-Player-Movements/tree/master/examples>

Urban/Highway Driving



LevelXData (rounD, exiD, hiD) [1]



Subdomain: Traffic Management

Captures urban settings from a top-down perspective, typically using RGB data. Focus is generally on developing intelligent mobility solutions.

WOMD [2]



Subdomain: Personal Driving

Captures urban/highway interactions from first-person view.

MAN Truck Scenes [3]



Subdomain: Freight

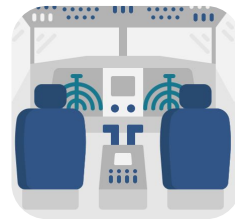
Focus on covering truck-specific surroundings (e.g., container terminals), larger local regions to understand the changes between chassis and cabin.

[1] LevelXData: <https://levelxdata.com>

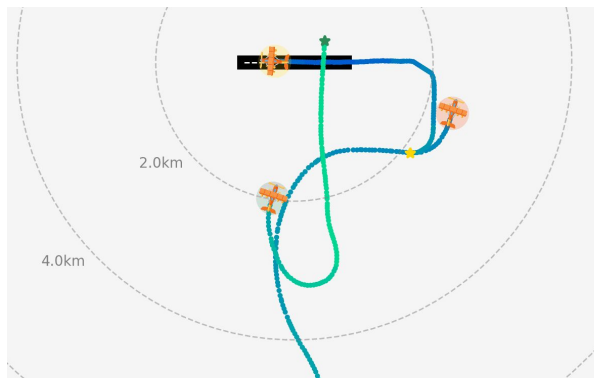
[2] Ettinger, S., Cheng, S., Caine, B., Liu, C., Zhao, H., Pradhan, S., ... & Anguelov, D. (2021). *Large scale interactive motion forecasting for autonomous driving: The waymo open motion dataset*. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 9710-9719).

[3] Fent, F., Kutenreich, F., Ruch, F., Rizwin, F., Juergens, S., Lechermann, L., ... & Lienkamp, M. (2024). Man trucks: A multimodal dataset for autonomous trucking in diverse conditions. *Advances in Neural Information Processing Systems*, 37, 62062-62082.

Aviation



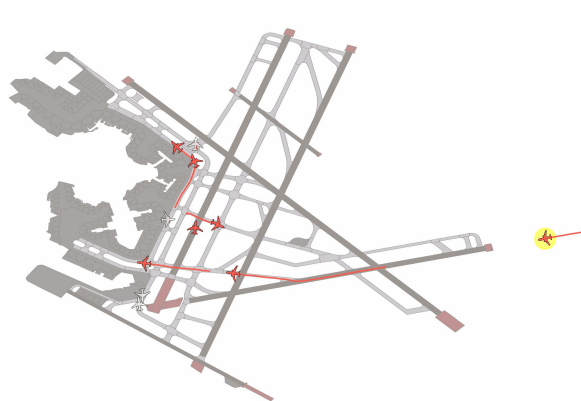
TrajAir [1]



Subdomain: Non-towered Airports

Captures aircraft interaction in a non-towered airport, i.e., airports where there is no centralized coordination via air traffic controllers. Interactions are centralized.

Amelia [2]



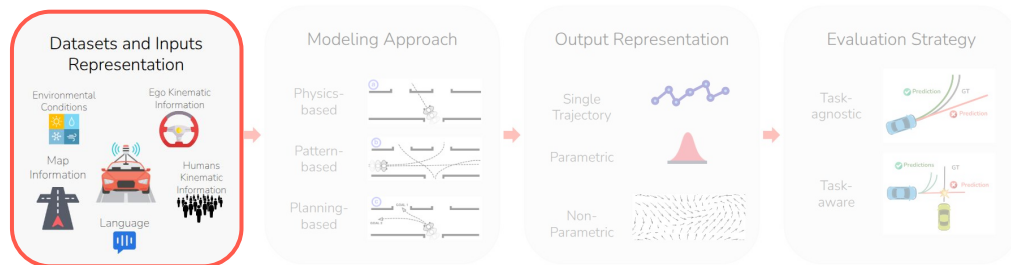
Subdomain: Airport Surface Movement Operations

Captures diverse airport operations in terminal airspace. Airports are primarily towered, meaning there is a centralized authority coordinating interactions.

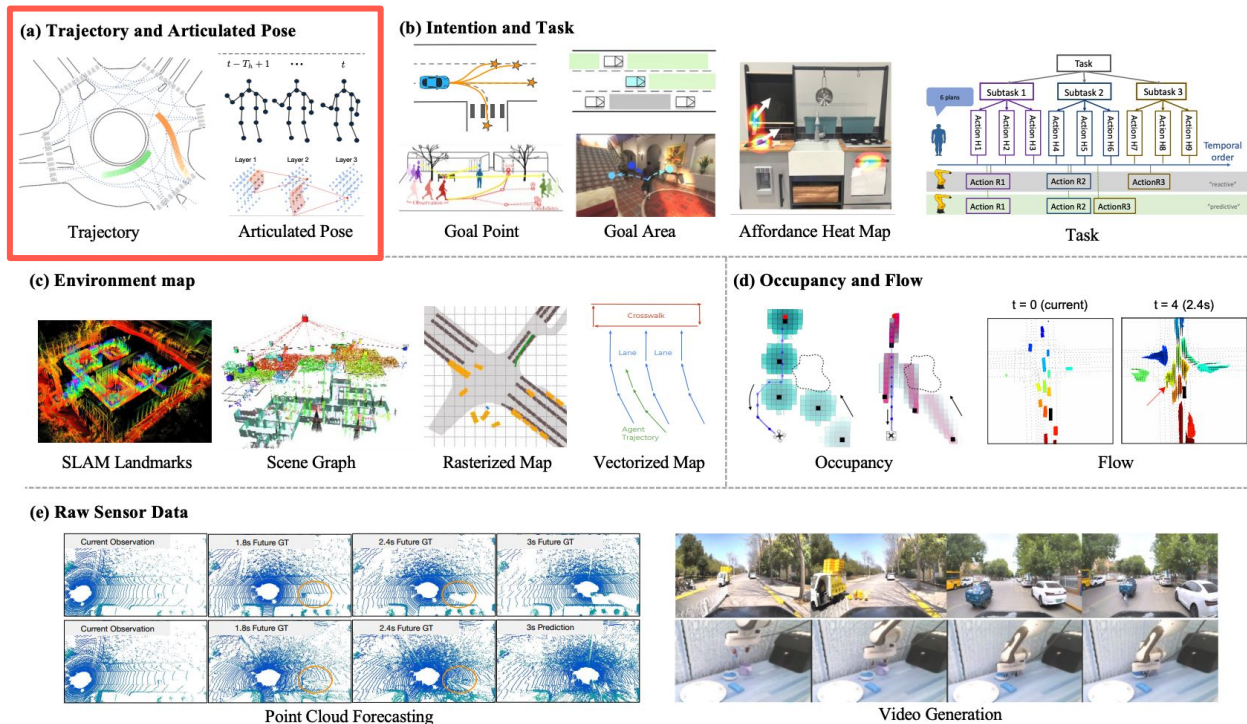
[1] Patrikar, J., Moon, B., Oh, J., & Scherer, S. (2022, May). Predicting like a pilot: Dataset and method to predict socially-aware aircraft trajectories in non-towered terminal airspace. In 2022 international conference on robotics and automation (icra) (pp. 2525-2531). IEEE.

[2] Navarro, I., Ortega, P., Patrikar, J., Wang, H., Ye, Z., Park, J. H., ... & Scherer, S. (2024). *AmeliaTF: A Large Model and Dataset for Airport Surface Movement Forecasting*. In AIAA AVIATION FORUM AND ASCEND 2024 (p. 4251).

Inputs Representation



Trajectories and Articulated Poses

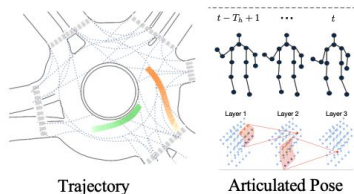


Trajectories. Motion is described as a sequence of 2D/3D waypoints. In driving settings, it often includes bounding box and agent type information.

Articulated Poses. Mostly relevant in human motion domains. Motion is represented as relative poses and joint angles.

Intention and Task

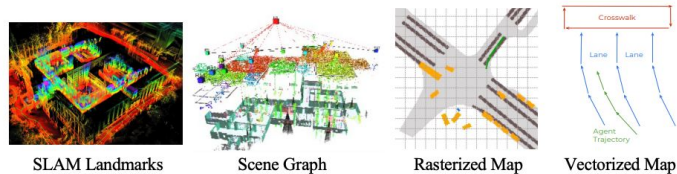
(a) Trajectory and Articulated Pose



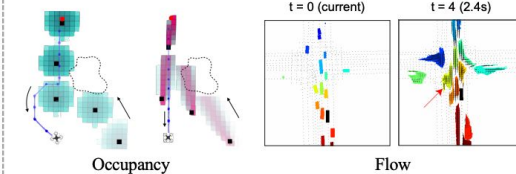
(b) Intention and Task



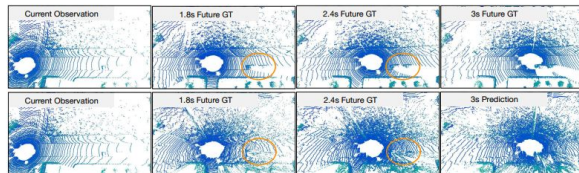
(c) Environment map



(d) Occupancy and Flow



(e) Raw Sensor Data



Point Cloud Forecasting

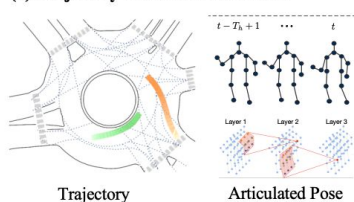


Video Generation

High-level inputs (and often outputs), representing goals, auxiliary tasks (e.g., identify relevant, causal information, dangerous areas, etc), task-specific skills.

Map

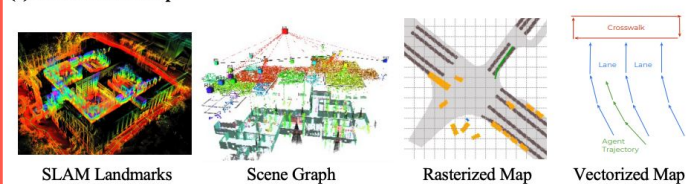
(a) Trajectory and Articulated Pose



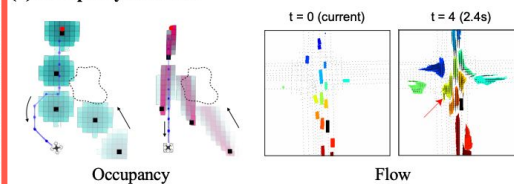
(b) Intention and Task



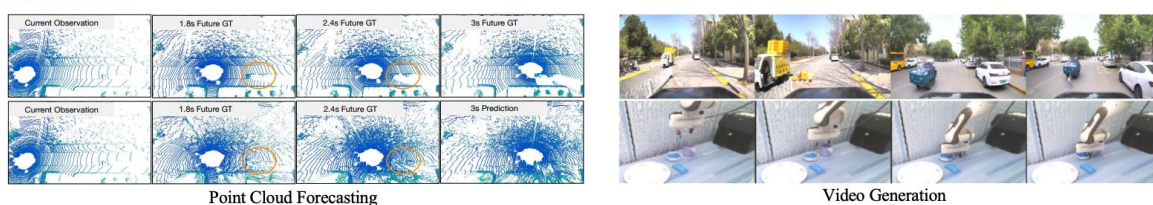
(c) Environment map



(d) Occupancy and Flow



(e) Raw Sensor Data



Landmarks. Maps can be represented as sparsely distributed points.

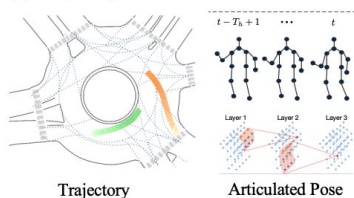
Scene graphs. High-level descriptions represented as hierarchical graphs that encode object-to-object relationships.

Raster maps. 2D birds-eye-view representation of a map depicting semantic areas. Useful for CNN-based predictive models.

Vectorized maps. Represents map (and agent) features as polylines. Better at representing complex topologies.

Occupancy and Flow

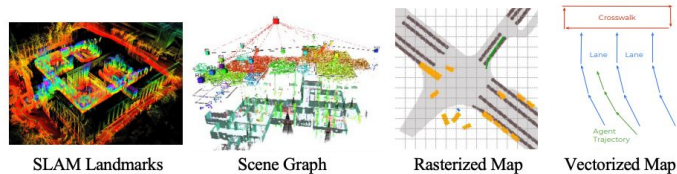
(a) Trajectory and Articulated Pose



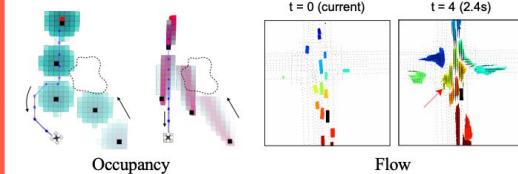
(b) Intention and Task



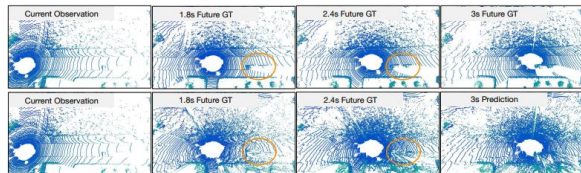
(c) Environment map



(d) Occupancy and Flow



(e) Raw Sensor Data



Point Cloud Forecasting

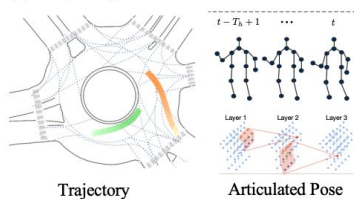


Video Generation

Scenes are represented as spatio-temporal grids and motion vectors, where each cell holds the probability that some agent is currently occupying it.

Raw Sensor Data

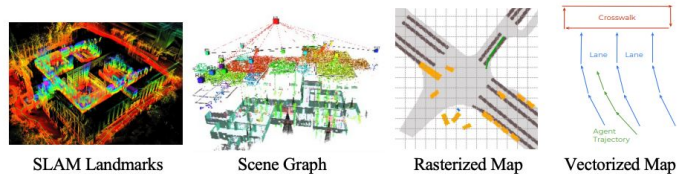
(a) Trajectory and Articulated Pose



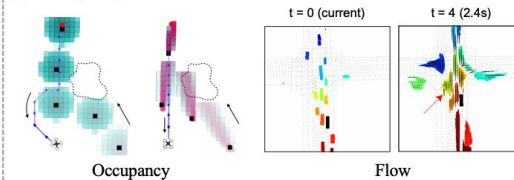
(b) Intention and Task



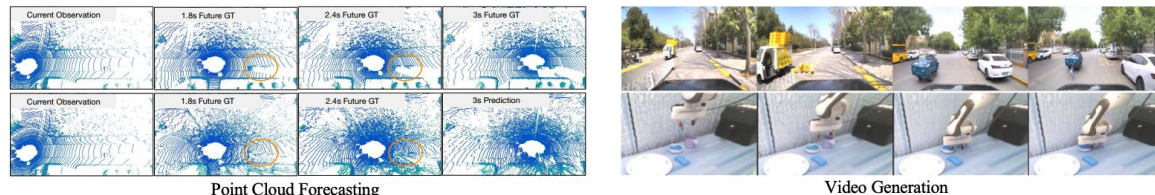
(c) Environment map



(d) Occupancy and Flow

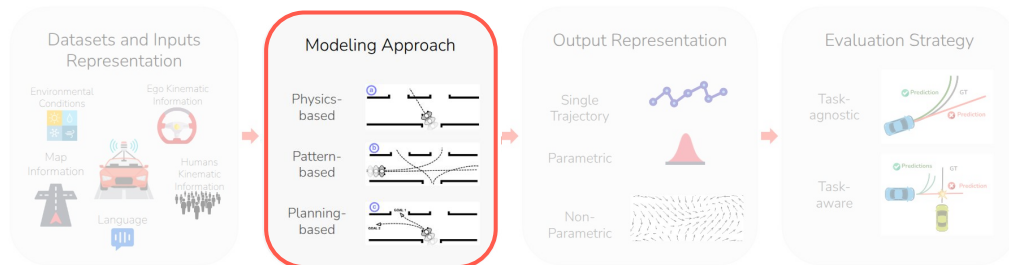


(e) Raw Sensor Data



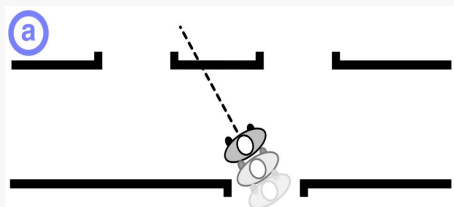
World model approaches that take features from upstream perception directly and predict in the same sensor format that they take in (e.g., RGB or LiDAR point clouds)

Modeling Approaches



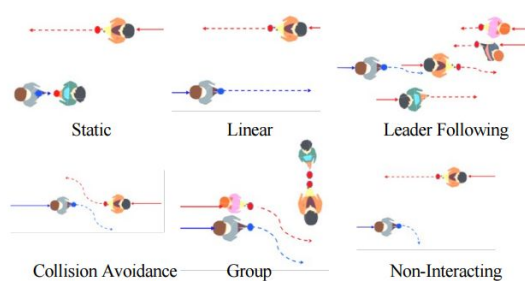
Physics-based Models

Motion is predicted by forward simulating a set of dynamics equations that follow a physics-inspired model.

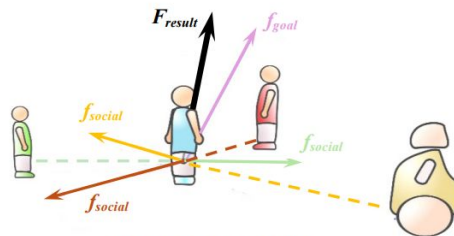


Generally easy to implement and acceptable under simple settings, such as small environments or short-horizon horizons predictions.

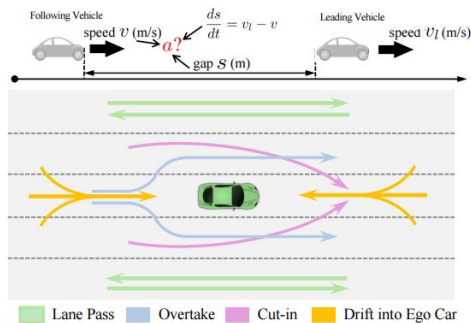
Difficult to effectively design hand-crafted models that account for all environmental and social complexities.



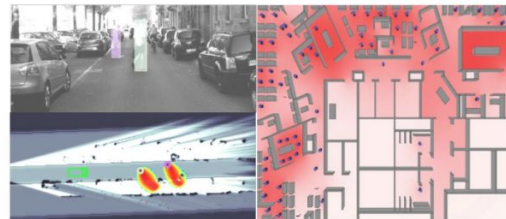
(a) Pedestrian Motion Patterns



(c) Social Force Model



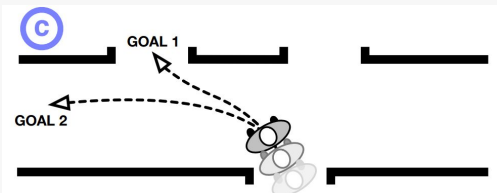
(b) Vehicle Motion Patterns



(d) Occupancy Probability

Planning-based Models

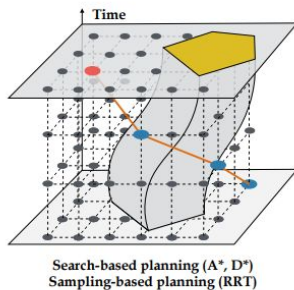
Motion is predicted by explicitly reasoning about the agent's long-term motion goals, and computing possible paths that attain those goals.



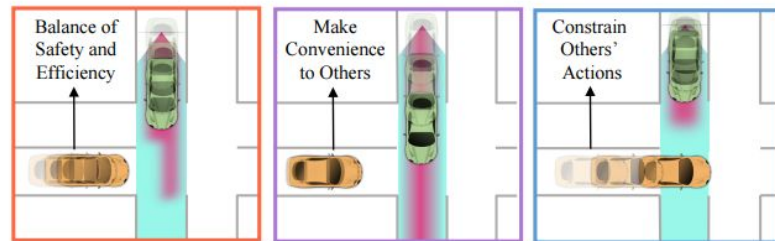
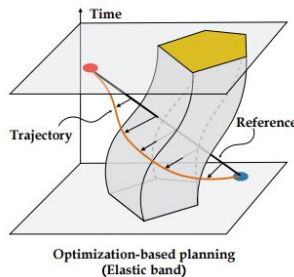
Goals, behavior heuristics and interactions can be modeled explicitly.

Assuming agents behave a certain way (e.g., rationally) can be suboptimal.

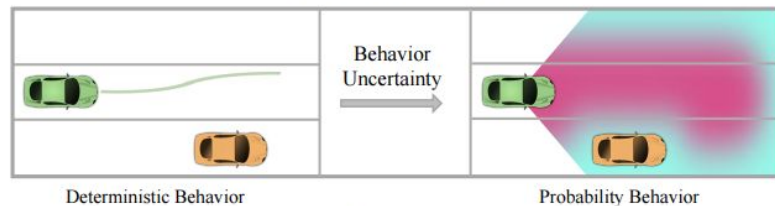
Planning by construction



Planning by modification



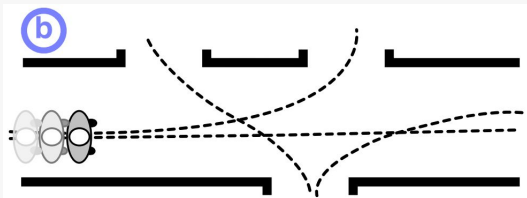
Interactive Planning



Probabilistic Planning

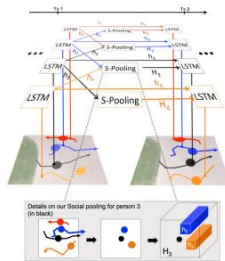
Pattern-based Models

Approximate a dynamics function from data, discovering statistical behavioral patterns.

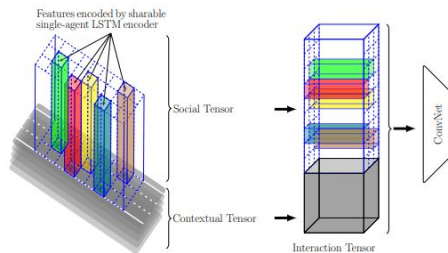


Can capture complex motion patterns directly from the data. More suitable for longer-horizon settings.

Generalizability to unseen settings is challenging.

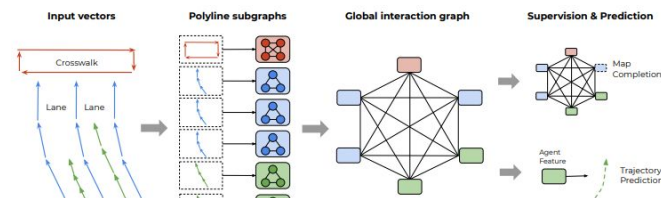


(a) RNN

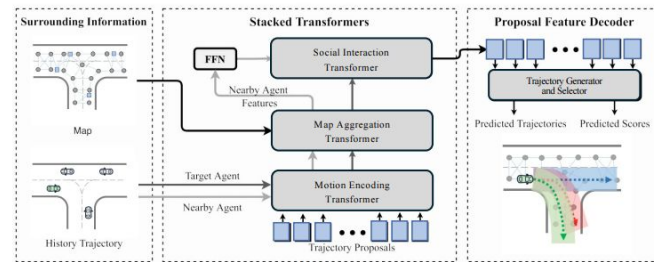


(b) CNN

Main focus of this lecture!

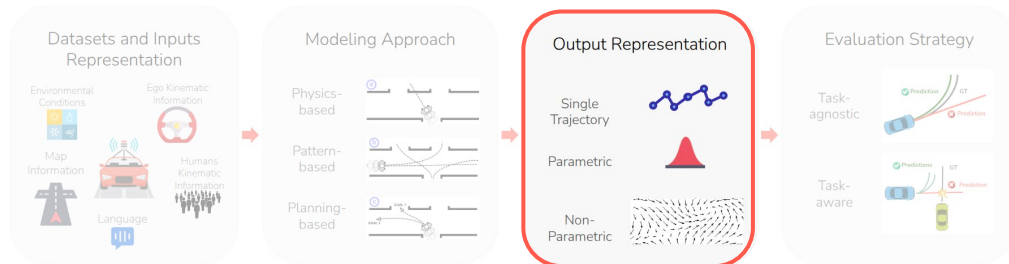


(c) GNN



(d) Attention

Output Representations



Types of Predictions

Single Trajectory

Model predicts a single, deterministic trajectory.



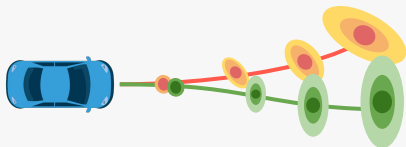
Simple to implement.

Cannot characterize multimodality and uncertainty.

Example: Constant Velocity Model

Parametric

Model outputs the parameters of a probability distribution, or mixture of distributions.



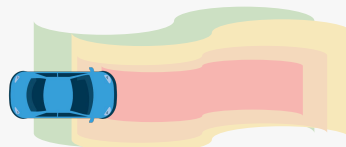
Captures prediction uncertainty, and mixtures can represent multiple possible futures.

Underlying distribution might not match the true distribution; fixed number of modes; potential collapse.

Examples: Social LSTM (Gaussian), Multipath (GMM over anchors), Trajectron++ (GMM output)

Non-Parametric

Model outputs without assuming a parametric distribution (e.g., occupancy, trajectory samples).



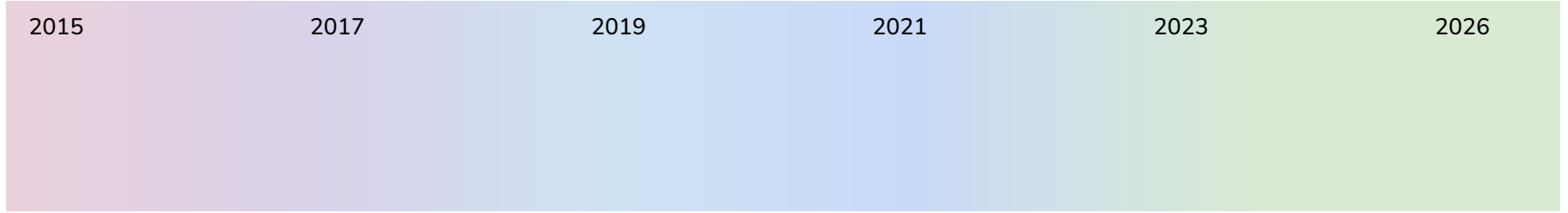
Flexible at capturing complex multimodal behaviors, with no fixed number of modes.

Often more difficult to evaluate, needing many samples; can be computationally expensive.

Examples: Social GAN, Diffusion, Occupancy Grids

Evolution of Pattern-based Approaches

Pattern-based Approaches Timeline



LSTMs + Local Social Pooling

2015

2017

2019

2021

2023

2026

Idea: Use recurrent sequence models (LSTMs) to model agent temporal dependencies and add a “social pooling” layer to let hidden states of nearby agents influence each other.

Representative Work: Social LSTM (Alahi et. al, CVPR 2016)

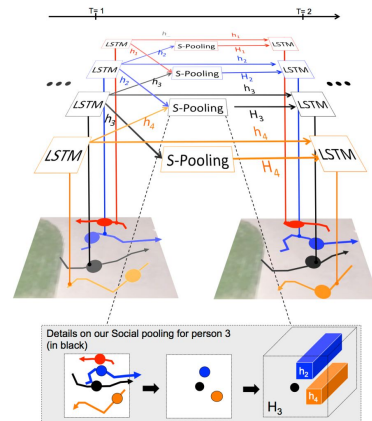
Influence: First type of strategy to replace hand-crafted rules with *learned* interactions; established a problem formulation that was widely explored by many follow-up works.

Strengths: Handle variable length histories; relatively simple and intuitive; strong performance in short-horizon settings.

Limitations: Unable to represent multiple futures; relied on hand-designed pooling.

Social LSTM: Human Trajectory Prediction in Crowded Spaces

Alexandre Alahi*, Krarthik Goel*, Vignesh Ramanathan, Alexandre Robicquet, Li Fei-Fei, Silvio Savarese
Stanford University



LSTMs + Global Social Pooling + GANs

2015

2017

2019

2021

2023

2026

Idea: Use GANs to produce multiple plausible futures but also train models adversarially to produce socially plausible trajectories.

Representative Work: Social GAN (Gupta et. al, CVPR 2018)

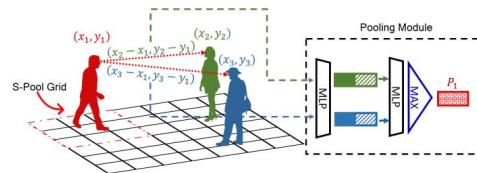
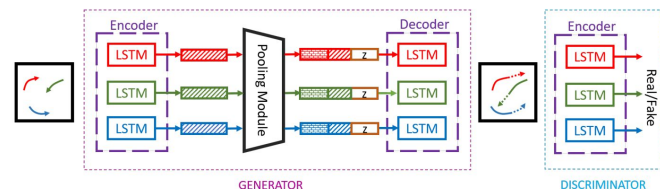
Influence: Predict multiple samples instead of unimodal representations.

Strengths: Sample diversity; global social pooling instead of k-closest.

Limitations: Difficult to train; evaluation of diversity/quality/realism was hard; no likelihood associated to a sample.

Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks

Agrim Gupta¹ Justin Johnson¹ Li Fei-Fei¹ Silvio Savarese¹ Alexandre Alahi^{1,2}
Stanford University¹ École Polytechnique Fédérale de Lausanne²



Structured Stochastic Models

2015 2017 2019 2021 2023 2026

Idea: Impose explicit structure and probabilistic prediction, through spatio-temporal graphs (agents=nodes, interactions=edges) or through predefined trajectory anchors. Predict multimodal distributions with latent-variable models (C-VAEs) and GMM outputs.

Representative Work: Trajectron (Ivanovic et. al, ICCV 2019), Multipath (Varadarajan et al, ICRA 2021).

Influence: Enabled conditioning on complex information via principled/structured representations; represent multimodality and uncertainty with explicit likelihoods.

Strengths: As above.

Limitations: Structure (graphs, anchors) is hand-designed; requires fixed number of modes; mostly per-agent predictions rather than joint ones.

The Trajectron: Probabilistic Multi-Agent Trajectory Modeling With Dynamic Spatiotemporal Graphs

Boris Ivanovic Marco Pavone
Stanford University

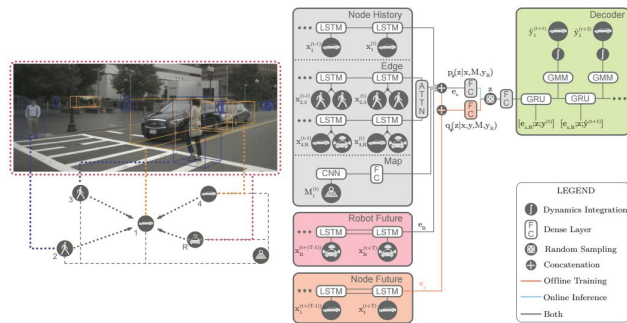


Figure is actually Trajectron++ (2020)

Transformer and Attention-based Models

2015

2017

2019

2021

2023

2026

Idea: Use transformers to jointly model social, temporal and other contextual relationships. Other attention-based models (e.g., GATs) were widely explored here.

Representative Work(s): AgentFormer (Yuan et al, ICCV 2021), SceneTransformer (Ngiam et al, ICLR 2021), MotionTransformer (Shi et al, NeurIPS 2022), Wayformer (Nayakanti et al, ICRA 2023), DAGNet (Monti et al, ICPR 2021) and many others.

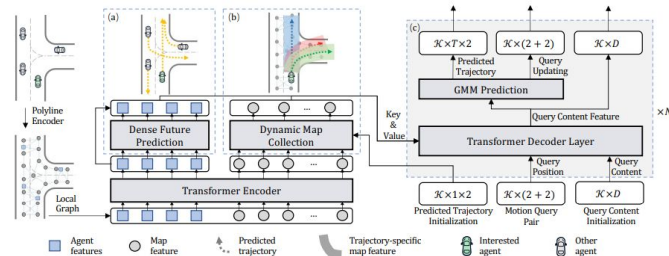
Influence: Attention enabled complex, long-range and joint modeling; works well with large-scale datasets.

Strengths: Parallelizable; design flexibility (easily compatible with deterministic and probabilistic models).

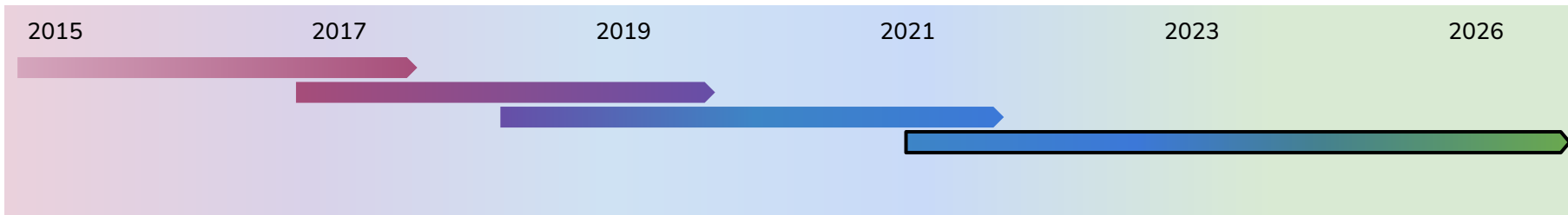
Limitations: Attention cost grows quadratically with the number of agents and map elements; generalization failures in unseen conditions.

Motion Transformer with Global Intention Localization and Local Movement Refinement

Shaoshuai Shi, Li Jiang, Dengxin Dai, Bernt Schiele
Max Planck Institute for Informatics, Saarland Informatics Campus
{ssh, lij, ddai, schiele}@mpi-inf.mpg.de



Transformer-based Models



Wayformer: Motion Forecasting via Simple & Efficient Attention Networks

Nigamaa Nayakanti*
nigamaa@waymo.com

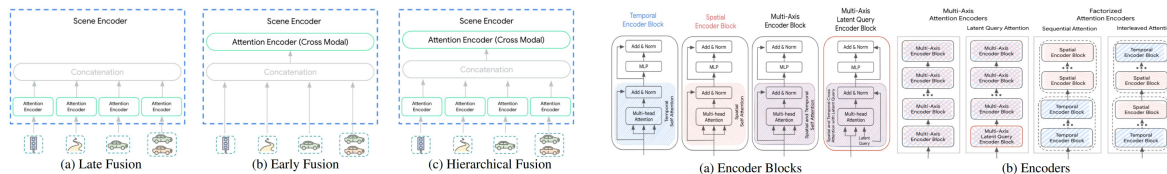
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Kratarth Goel
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Khaled S. Refaat
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Benjamin Sapp
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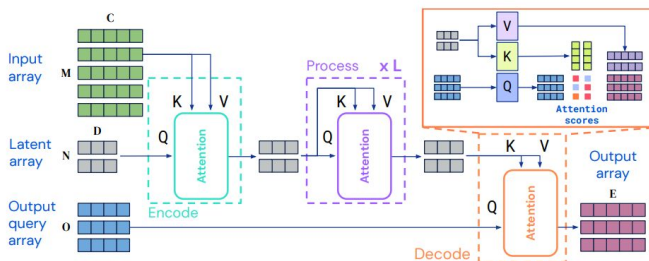
PERCEIVER IO: A GENERAL ARCHITECTURE FOR STRUCTURED INPUTS & OUTPUTS

Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, Carl Doersch, Catalin Ionescu,

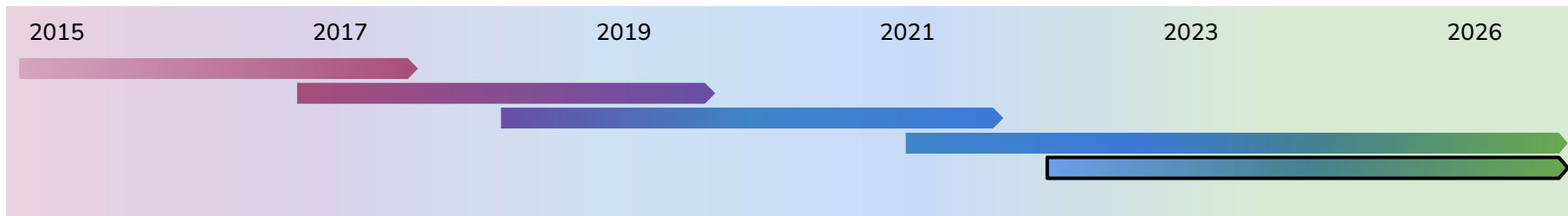
David Ding, Skanda Koppula, Daniel Zoran, Andrew Brock, Evan Shelhamer, Olivier Hénaff,

Matthew M. Botvinick, Andrew Zisserman, Oriol Vinyals, João Carreira

DeepMind



Diffusion-based Models



Idea: Learn joint distributions by gradually denoising samples.

Representative Work: MotionDiffuser (Jiang et al., CVPR 2023) and SceneDiffuser (Jiang et al, NeurIPS, 2024) .

Influence: Guidance and inpainting enabled controllable generation, allowing a single model to handle prediction, simulation, and scenario generation.

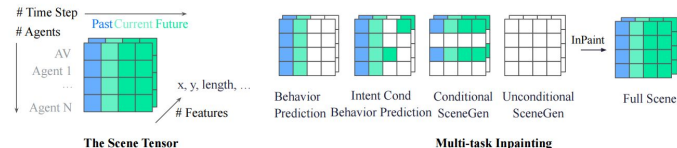
Strengths: Strong sample diversity and realism, stable training with less mode collapse than GANs; no fixed number of modes.

Limitations: High inference cost from iterative denoising; no tractable likelihood, making evaluation hard.

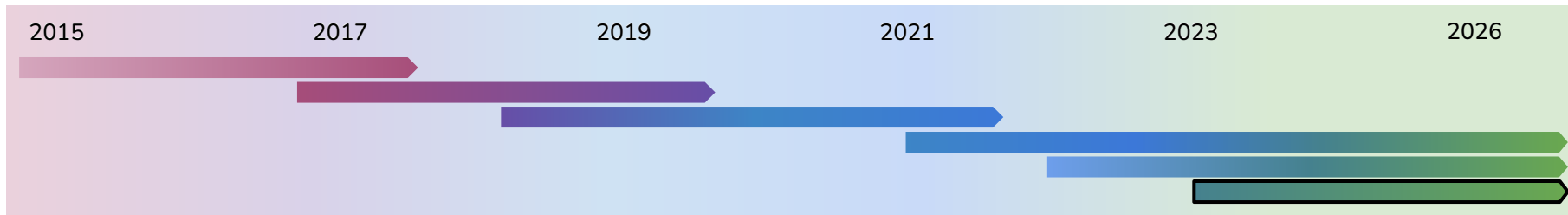
SceneDiffuser: Efficient and Controllable Driving Simulation Initialization and Rollout

Chiyu Max Jiang Yijing Bai* Andre Cornman* Christopher Davis* Xiukun Huang*
Hong Jeon* Sakshum Kulshrestha* John Lambert* Shuangyu Li* Xuanyu Zhou*
Carlos Fuertes Chang Yuan Mingxing Tan Yin Zhou Dragomir Anguelov

Waymo LLC



Trajectory Forecasting + Large Models



Idea: Use LLMs/VLMs to add reasoning, scene understanding, and language interfaces to trajectory forecasting, either as priors, as direct predictors, or as data generators.

Representative Work: LMTraj (Bae et al., CVPR 2024)

Influence: Providing predictors with commonsense and open-world reasoning.

Strengths: Zero- and few-shot reasoning, flexible conditioning through prompts, and human-readable reasoning.

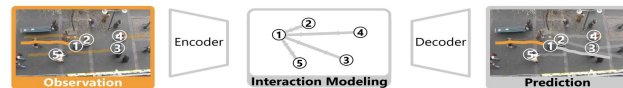
Limitations: High latency for real-time use; prone to hallucinations; weak at precise spatial and numerical reasoning.

Can Language Beat Numerical Regression? Language-Based Multimodal Trajectory Prediction

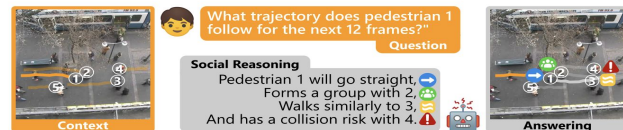
Inhwan Bae¹, Junoh Lee² and Hae-Gon Jeon^{1,2*}

¹AI Graduate School, ²School of Electrical Engineering and Computer Science
Gwangju Institute of Science and Technology, Gwangju, South Korea

{inhwanbae, junoh}@gm.gist.ac.kr, haegonj@gist.ac.kr

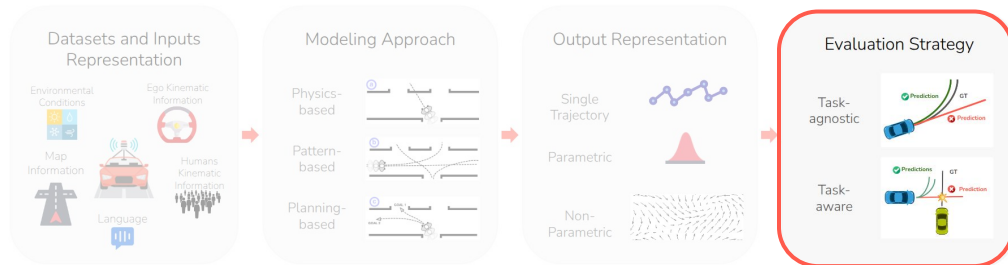


(a) Previous trajectory prediction



(b) Our language-based trajectory prediction

Evaluation Strategies

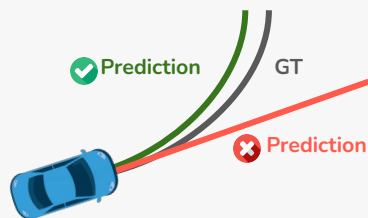


Task-Agnostic Evaluation

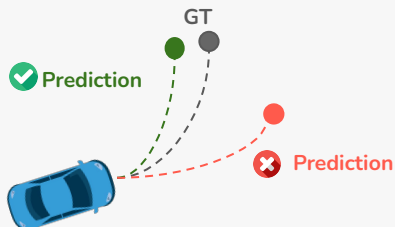
Metrics that focus on quantifying the similarity between the predicted and ground truth motion, but also capture the multi-modality of human behavior.

Accuracy / Realism

Minimum Average Displacement Error (mADE)

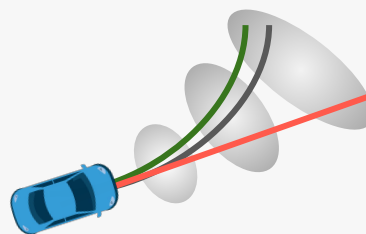


Minimum Final Displacement Error (mFDE)



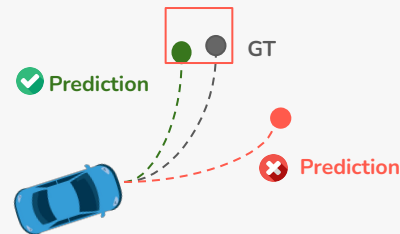
Uncertainty

Kernel Density Estimate (KDE)-based NLL



Coverage

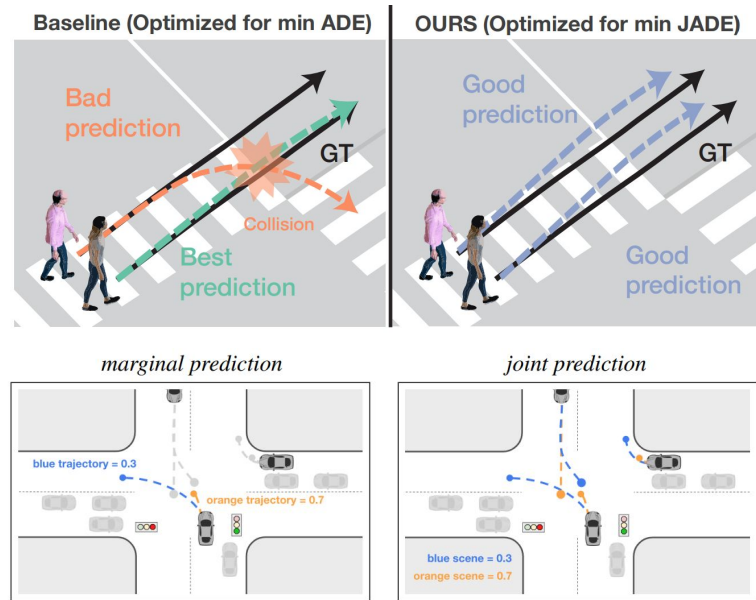
Mean Average Precision (mAP)



Task-Agnostic Evaluation

Single-agent metrics do not capture **joint performance**, which can lead to unnatural and inconsistent social behavior, e.g.;

- Colliding trajectories
- Diverging trajectories for social groups



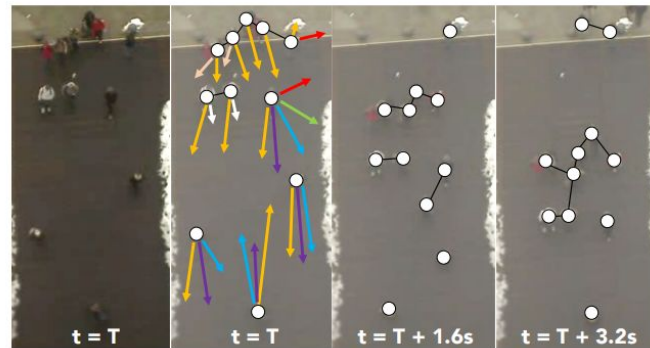
Sources: top figure [1], bottom figure [2]

[1] Weng, E., Hoshino, H., Ramanan, D., & Kitani, K. (2023). *Joint metrics matter: A better standard for trajectory forecasting*. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 20315-20326).

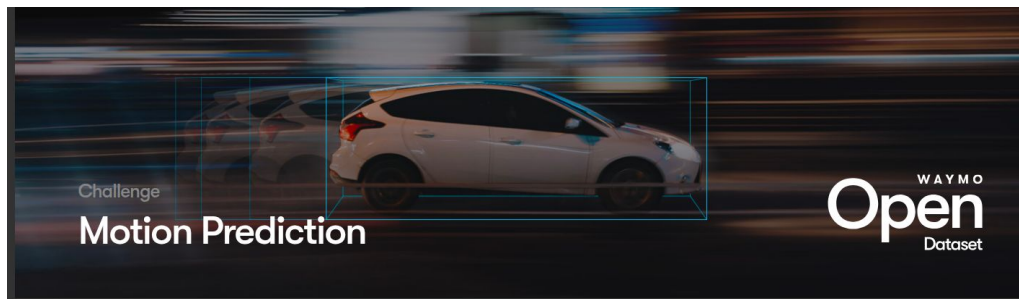
[2] Ngiam, J., Caine, B., Vasudevan, V., Zhang, Z., Chiang, H. T. L., Ling, J., ... & Shlens, J. (2021). *Scene transformer: A unified architecture for predicting multiple agent trajectories*. arXiv preprint arXiv:2106.08417.

Benchmarks: ETH / UCY (Human Crowds)

Method	ADE ₂₀ /FDE ₂₀ ↓ (m), $K = 20$ Samples					Average
	ETH	Hotel	Univ	Zara1	Zara2	
SGAN [15]	0.81/1.52	0.72/1.61	0.60/1.26	0.34/0.69	0.42/0.84	0.58/1.18
SoPhie [44]	0.70/1.43	0.76/1.67	0.54/1.24	0.30/0.63	0.38/0.78	0.54/1.15
Transformer-TF [12]	0.61/1.12	0.18/0.30	0.35/0.65	0.22/0.38	0.17/0.32	0.31/0.55
STAR [55]	0.36/0.65	0.17/0.36	0.31/0.62	0.26/0.55	0.22/0.46	0.26/0.53
PECNet [34]	0.54/0.87	0.18/0.24	0.35/0.60	0.22/0.39	0.17/0.30	0.29/0.48
Trajectron++ [45]	0.39/0.83	0.12/0.21	0.20/0.44	0.15/0.33	0.11/0.25	0.19/0.41
Ours (AgentFormer)	0.45/ 0.75	0.14/0.22	0.25/0.45	0.18/ 0.30	0.14/ 0.24	0.23/ 0.39



Benchmarks: WOMD, nu-Scenes, Argoverse (Urban Driving)



Method Name	Object type	Evaluation time	Soft mAP v2	mAP v2	minADE	minFDE	Miss rate	Overlap Rate	Uses public model pretraining
	All	Avg	Show results well						
MIRv3	All	Avg	0.4967	0.4859	0.5554	1.1062	0.1098	0.1279	
IMPACT	All	Avg	0.4801	0.4598	0.5563	1.1295	0.1087	0.1258	
RMP_Ensemble	All	Avg	0.4737	0.4531	0.5564	1.1188	0.1084	0.1259	
ModelSeq	All	Avg	0.4737	0.4665	0.5680	1.1766	0.1204	0.1275	
HelioFusion	All	Avg	0.4732	0.4624	0.5690	1.1596	0.1183	0.1243	
TranXw	All	Avg	0.4710	0.4604	0.5714	1.1667	0.1162	0.1272	
BeTop	All	Avg	0.4698	0.4587	0.5716	1.1668	0.1183	0.1272	
BehaveOcc	All	Avg	0.4678	0.4566	0.5723	1.1668	0.1176	0.1278	
RMP-YOLO	All	Avg	0.4673	0.4523	0.5737	1.1697	0.1160	0.1266	
QMTR	All	Avg	0.4649	0.4445	0.5702	1.1627	0.1177	0.1269	
UniMotion	All	Avg	0.4648	0.4540	0.5693	1.1574	0.1156	0.1266	

nuScenes prediction task

Leaderboard

Search:

Export as JSON

	Date	Name	MinADE_5	MinADE_10	MissRateTopK_2_5	MissRateTopK_2_10	MinFDE_1	OffRoadRate
>	2022-05-25	DGCN_ST_LANE	1.092	0.974	46.99%	42.22%	3.624	0.124
>	2024-04-12	CASPFormer	1.148	1.148	47.83%	47.83%	6.702	0.011
>	2022-05-01	DSS	1.192	0.887	49.23%	30.99%	6.640	0.018
>	2022-09-06	Jack	1.211	0.894	56.65%	33.38%	7.502	0.019
>	2022-05-19	LaPred++	1.235	1.054	52.60%	46.19%	7.580	0.060
>	2022-03-07	PGP	1.267	0.945	51.90%	34.34%	7.171	0.028
>	2022-01-24	CASPNet_v2	1.275	1.010	53.15%	31.81%	7.023	0.011
>	2021-11-17	THOMAS	1.325	1.041	55.10%	42.40%	6.712	0.031
>	2022-01-23	JAL-MTP	1.356	0.988	54.88%	35.75%	7.870	0.025
>	2021-09-15	Autobot	1.368	1.026	62.23%	43.87%	8.190	0.025



Argoverse 2: Motion Forecasting Competition

Organized by: [argoverse](#)

Starts on: Apr 17, 2022 8:00:00 PM EST (GMT - 4:00)

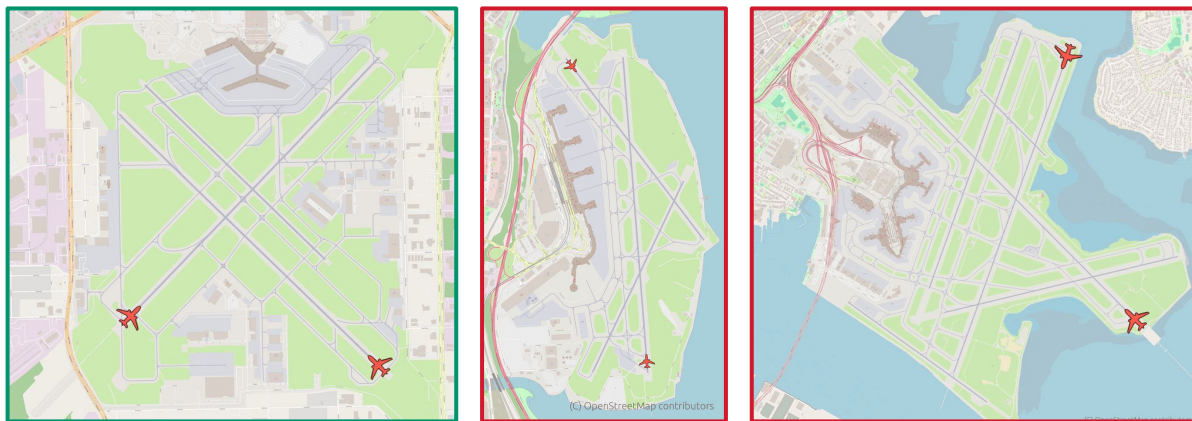
Ends on: May 31, 2022 7:59:59 PM EST (GMT - 4:00)

★ 237

[Overview](#) [Evaluation](#) [Phases](#) [Participate](#) [Leaderboard](#)

Benchmarks: Amelia-10 (Aviation)

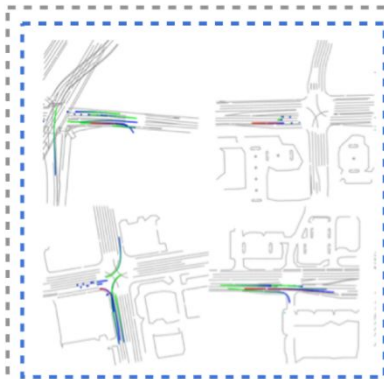
How can we design and evaluate models that generalize across environment topologies?



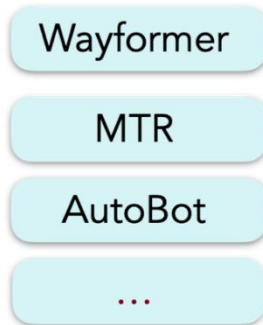
Benchmarks: UniTraj



Raw Benchmarks



1. Unified Datasets



2. Unified Models



3. Unified Evaluations

UniTraj



Train



Test

RQ1: Generalization



Train



Test

RQ2: Data Scaling



Train

Test

Dataset Analysis



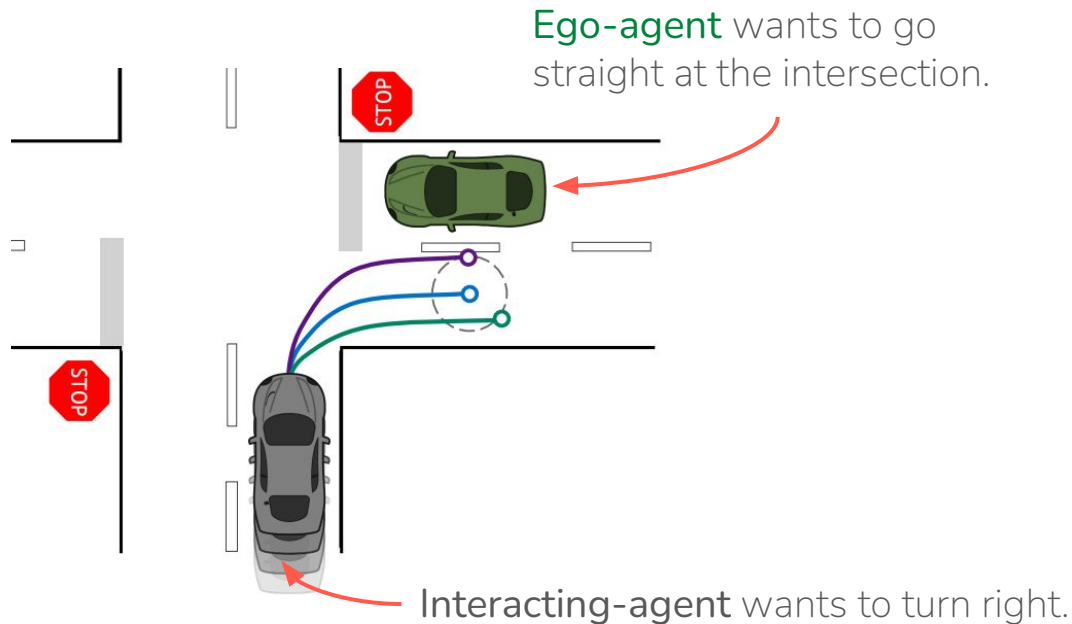
...

Problems with Task-Agnostic Evaluation

The **blue trajectory** is the ground truth

The **green** and **purple** trajectories are the ego-agent's predictions.

Imagine that both were predicted to be equally likely, but the purple one has lower ADE.

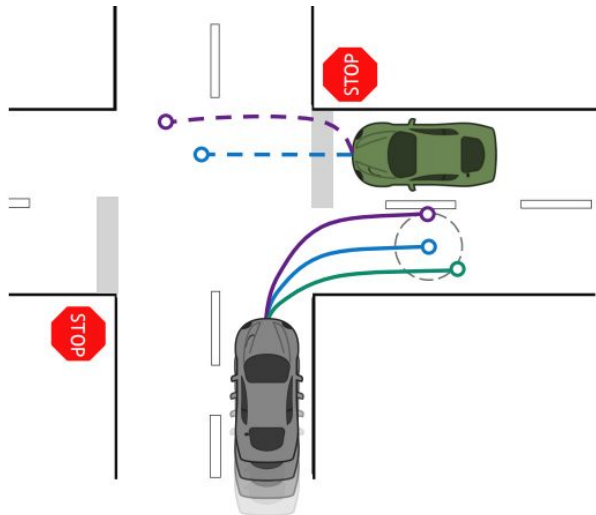


Problems with Task-Agnostic Evaluation

The **blue trajectory** is the ground truth

The **green** and **purple** trajectories are the ego-agent's predictions.

Imagine that both were predicted to be equally likely, but the purple one has lower ADE.



The **purple** prediction is used to inform the ego-agent.

The ego realizes it is a potentially dangerous trajectory.

Thus it takes an evasive maneuver, which incurs a higher planning cost than the **ground truth plan**.

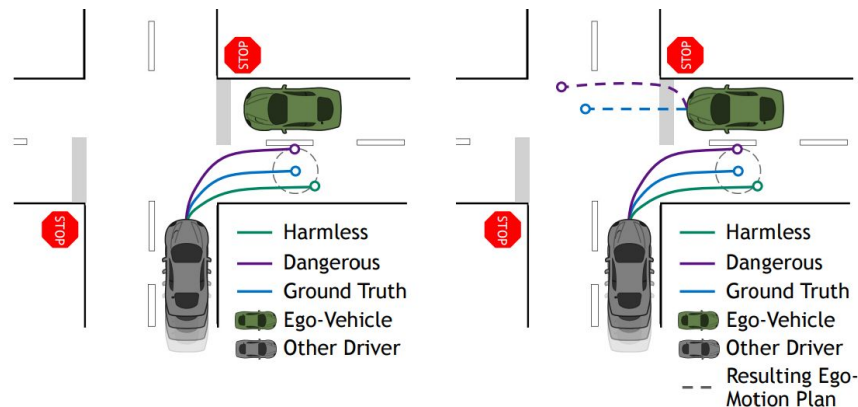
Problems with Task-Agnostic Evaluation

Task-agnostic metrics are disconnected from the downstream task and from real-world evaluation and deployment.

Task-Aware Evaluation

Focus is on evaluating and addressing:

- Prediction under perception uncertainty [1].
- Implications of prediction failures on downstream tasks [2, 3, 4].



[1] Stoler, B., Jana, M., Hwang, S., & Oh, J. (2023, October). *T2FPV: Dataset and method for correcting first-person view errors in pedestrian trajectory prediction*. In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 4037-4044). IEEE.

[2] Ivanovic, B., & Pavone, M. (2021). *Rethinking trajectory forecasting evaluation*. arXiv preprint arXiv:2107.10297.

[3] Farid, A., Veer, S., Ivanovic, B., Leung, K., & Pavone, M. (2023, March). *Task-relevant failure detection for trajectory predictors in autonomous vehicles*. In Conference on Robot Learning (pp. 1959-1969). PMLR.

[4] Nakamura, K., Tian, T., & Bajcsy, A. (2025, January). *Not All Errors Are Made Equal: A Regret Metric for Detecting System-level Trajectory Prediction Failures*. In Conference on Robot Learning (pp. 4051-4065). PMLR.

Open Challenges in Motion Prediction

The State-of-the-Art

Motion Transformer (MTR), 1st Place

Waymo Open Motion Prediction

Challenge, 2022.

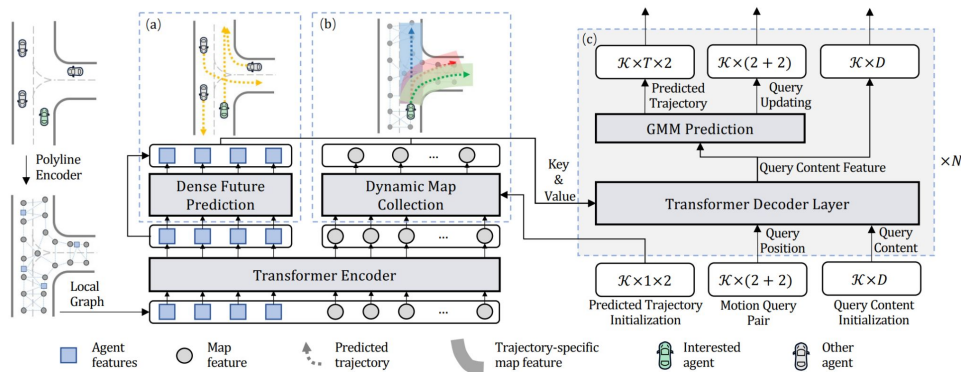


Table 1: Performance comparison of marginal motion prediction on the validation and test set of Waymo Open Motion Dataset. †: The results are shown in *italic* for reference since their performance is achieved with model ensemble techniques. We only evaluate our default setting MTR on the test set by submitting to official test server due to the limitation of submission times of WOMD.

	Method	Reference	minADE ↓	minFDE ↓	Miss Rate ↓	mAP ↑
Test	MotionCNN [26]	CVPRw 2021	0.7400	1.4936	0.2091	0.2136
	ReCoAt [66]	CVPRw 2021	0.7703	1.6668	0.2437	0.2711
	DenseTNT [21]	ICCV 2021	1.0387	1.5514	0.1573	0.3281
	SceneTransformer [37]	ICLR 2022	0.6117	1.2116	0.1564	0.2788
	MTR (Ours)	-	0.6050	1.2207	0.1351	0.4129
	† MultiPath++ [49]	ICRA 2022	<i>0.5557</i>	<i>1.1577</i>	<i>0.1340</i>	<i>0.4092</i>
	† MTR-Advanced-ens (Ours)	-	<i>0.5640</i>	<i>1.1344</i>	<i>0.1160</i>	<i>0.4492</i>

The State-of-the-Art



Is the problem solved?

Waymo Open Motion
Prediction Challenge
leaderboard, 2024.

Method Name	Object type	Evaluation time	Soft mAP v2	mAP v2	minADE	minFDE	Miss rate	Overlap Rate
	All	Avg	Show results with					
MTR v3	All	Avg	0.4967	0.4859	0.5554	1.1062	0.1098	0.1279
IMPACT	All	Avg	0.4801	0.4598	0.5563	1.1295	0.1087	0.1258
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QMTR-V2	All	Avg	0.4646	0.4441	0.5700	1.1621	0.1174	0.1270

Variants of MTR

Evaluation metrics have not
improved significantly since

Benchmark–Reality Gap



The Reality:

Human-involved rear-end accidents

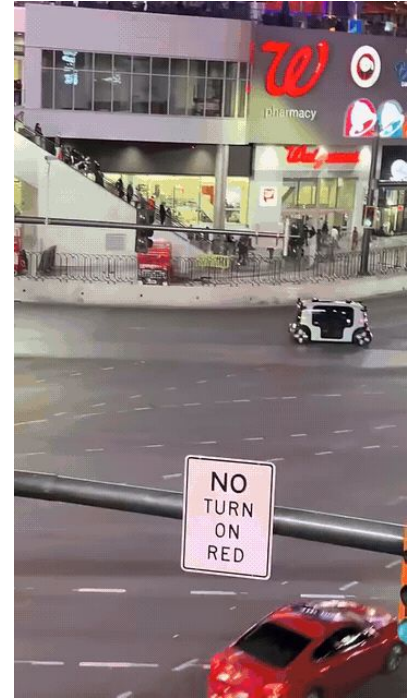
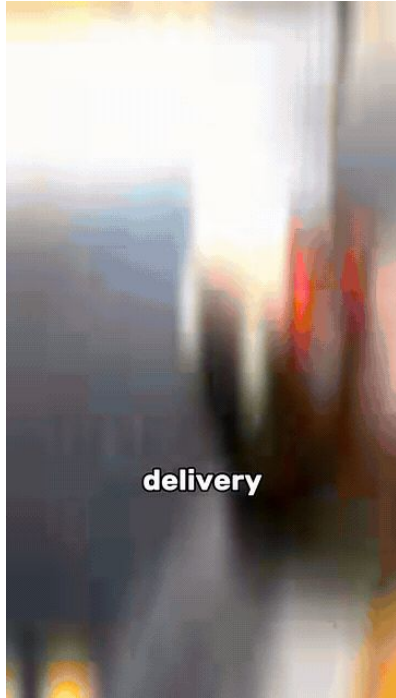
1.9 per million miles traveled

vs.

ADS-involved rear-end accidents

9.1 per million miles traveled

Even within their operational domains, they're not fully reliable



What's behind the generalization gap?

Existing challenges in the field:

Perception-level

Perception errors and perturbations to the input data

Data-level

Non-uniform data coverage

Control-level

How do we use forecasts downstream?

Operational Restrictions

Ensuring uniform data coverage is difficult and expensive to attain, which makes generalizability **in unseen conditions** challenging.

Operational Restrictions:



Geofences



Weather



Time of Day

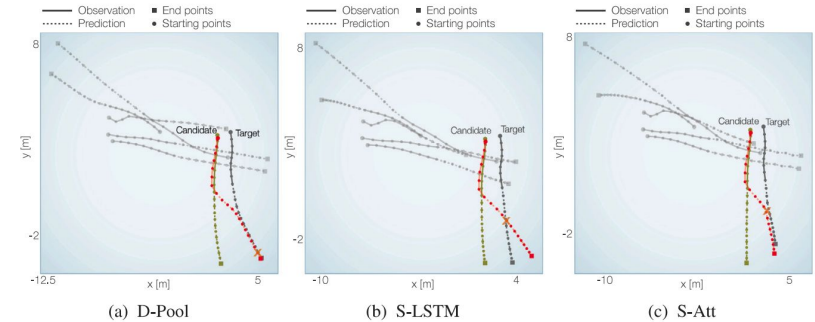
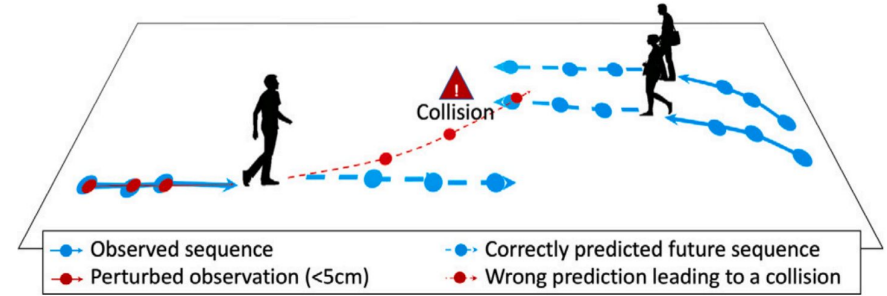


Road Type



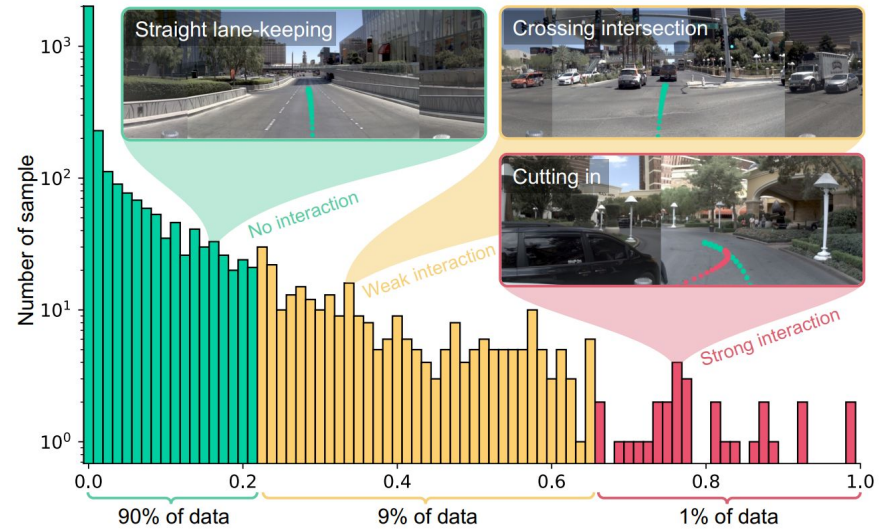
Brittleness to Perception Errors

Carefully-crafted perturbations can cause significant prediction failures that lead to unrealistic and/or unsafe behavior.



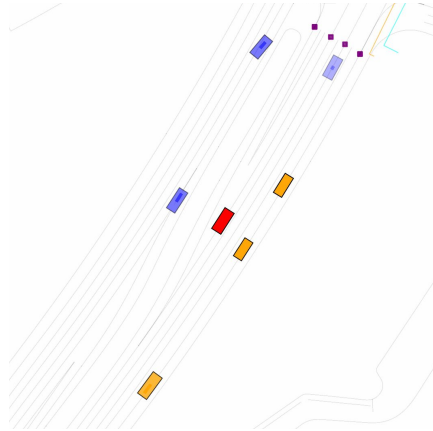
Lack of Meaningful Interactivity

The majority of recorded driving data consists of uneventful driving with limited interactions [1].



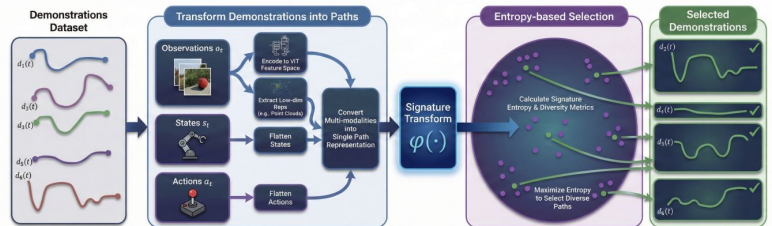
Lack of Meaningful Interactivity?

*The majority of recorded driving data consists of **uneventful** driving with limited interactions [1].*

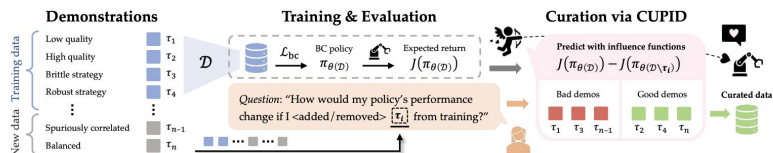


Selecting better samples for training

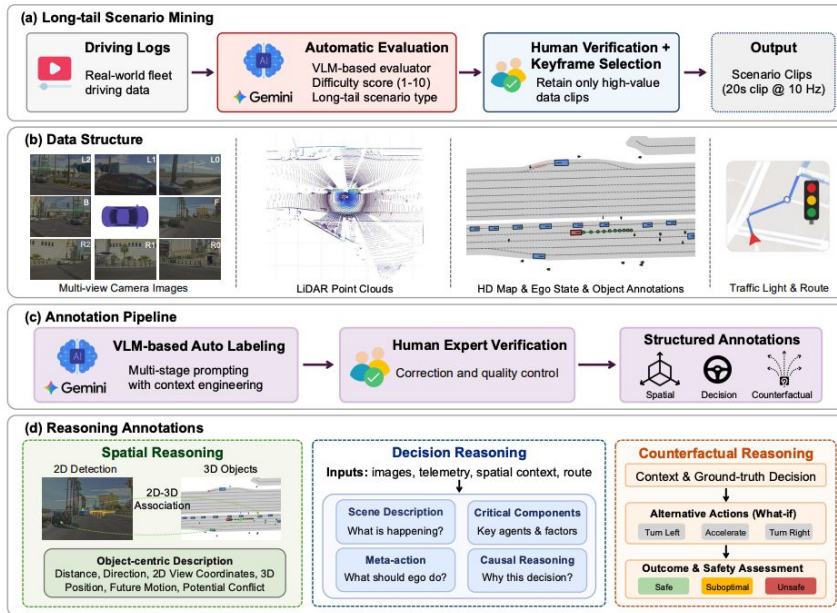
Diversity You Can Actually Measure: A Fast, Model-Free Diversity Metric for Robotics Datasets



CUPID: Curating Data your Robot Loves with Influence Functions



nuReasoning: A Reasoning-Centric Dataset and Benchmark for Long-Tail Autonomous Driving



Recommended Papers: Fundamentals, Datasets, Models

Surveys and Fundamentals:

- ❑ Wang, L., Lavoie, M. A., Papais, S., Nisar, B., Chen, Y., Ding, W., ... & Waslander, S. (2025). *Deployable and Generalizable Motion Prediction: Taxonomy, Open Challenges and Future Directions*. arXiv preprint arXiv:2505.09074.
- ❑ Rudenko, A., Palmieri, L., Herman, M., Kitani, K. M., Gavrila, D. M., & Arras, K. O. (2020). *Human motion trajectory prediction: A survey*. The International Journal of Robotics Research, 39(8), 895–935.

Models:

- ❑ Alahi, A., Goel, K., Ramanathan, V., Robicquet, A., Fei-Fei, L., & Savarese, S. (2016). *Social lstm: Human trajectory prediction in crowded spaces*. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 961–971).
- ❑ Yuan, Y., Weng, X., Ou, Y., & Kitani, K. M. (2021). *Agentformer: Agent-aware transformers for socio-temporal multi-agent forecasting*. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 9813–9823).
- ❑ Ngiam, J., Caine, B., Vasudevan, V., Zhang, Z., Chiang, H. T. L., Ling, J., ... & Shlens, J. (2021). *Scene transformer: A unified architecture for predicting multiple agent trajectories*. arXiv preprint arXiv:2106.08417.
- ❑ Gao, J., Sun, C., Zhao, H., Shen, Y., Anguelov, D., Li, C., & Schmid, C. (2020). *Vectornet: Encoding hd maps and agent dynamics from vectorized representation*. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 11525–11533).
- ❑ Shi, S., Jiang, L., Dai, D., & Schiele, B. (2022). *Motion transformer with global intention localization and local movement refinement*. Advances in Neural Information Processing Systems, 35, 6531–6543.
- ❑ Nayakanti, N., Al-Rfou, R., Zhou, A., Goel, K., Refaat, K. S., & Sapp, B. (2022). *Wayformer: Motion forecasting via simple & efficient attention networks*. arXiv preprint arXiv:2207.05844.
- ❑ Feng, L., Bahari, M., Amor, K. M. B., Zablocki, É., Cord, M., & Alahi, A. (2024, September). *Unitraj: A unified framework for scalable vehicle trajectory prediction*. In European Conference on Computer Vision (pp. 106–123). Cham: Springer Nature Switzerland.

Datasets:

- ❑ Chang, M. F., Lambert, J., Sangkloy, P., Singh, J., Bak, S., Hartnett, A., ... & Hays, J. (2019). *Argoverse: 3d tracking and forecasting with rich maps*. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 8748–8757).
- ❑ Ettinger, S., Cheng, S., Caine, B., Liu, C., Zhao, H., Pradhan, S., ... & Anguelov, D. (2021). *Large scale interactive motion forecasting for autonomous driving: The waymo open motion dataset*. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 9710–9719).
- ❑ Navarro, I., Ortega, P., Patrikar, J., Wang, H., Ye, Z., Park, J. H., ... & Scherer, S. (2024). *AmeliaTF: A Large Model and Dataset for Airport Surface Movement Forecasting*. In AIAA AVIATION FORUM AND ASCEND 2024 (p. 4251).

Evaluation:

- ❑ Ivanovic, B., & Pavone, M. (2021). *Rethinking trajectory forecasting evaluation*. arXiv preprint arXiv:2107.10297.
- ❑ Li, L., Lin, X., Huang, Y., Zhang, Z., & Hu, J. F. (2024). *Beyond minimum-of-N: Rethinking the evaluation and methods of pedestrian trajectory prediction*. IEEE Transactions on Circuits and Systems for Video Technology.
- ❑ Weng, E., Hoshino, H., Ramanan, D., & Kitani, K. (2023). *Joint metrics matter: A better standard for trajectory forecasting*. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 20315–20326).

Recommended Papers: Related / Frontier Topics

Scenario Generation, Re-simulation and Closed-loop Evaluation

- ❑ Suo, S., Wong, K., Xu, J., Tu, J., Cui, A., Casas, S., & Urtasun, R. (2023). *Mixsim: A hierarchical framework for mixed reality traffic simulation*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 9622–9631).
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Thank you!